

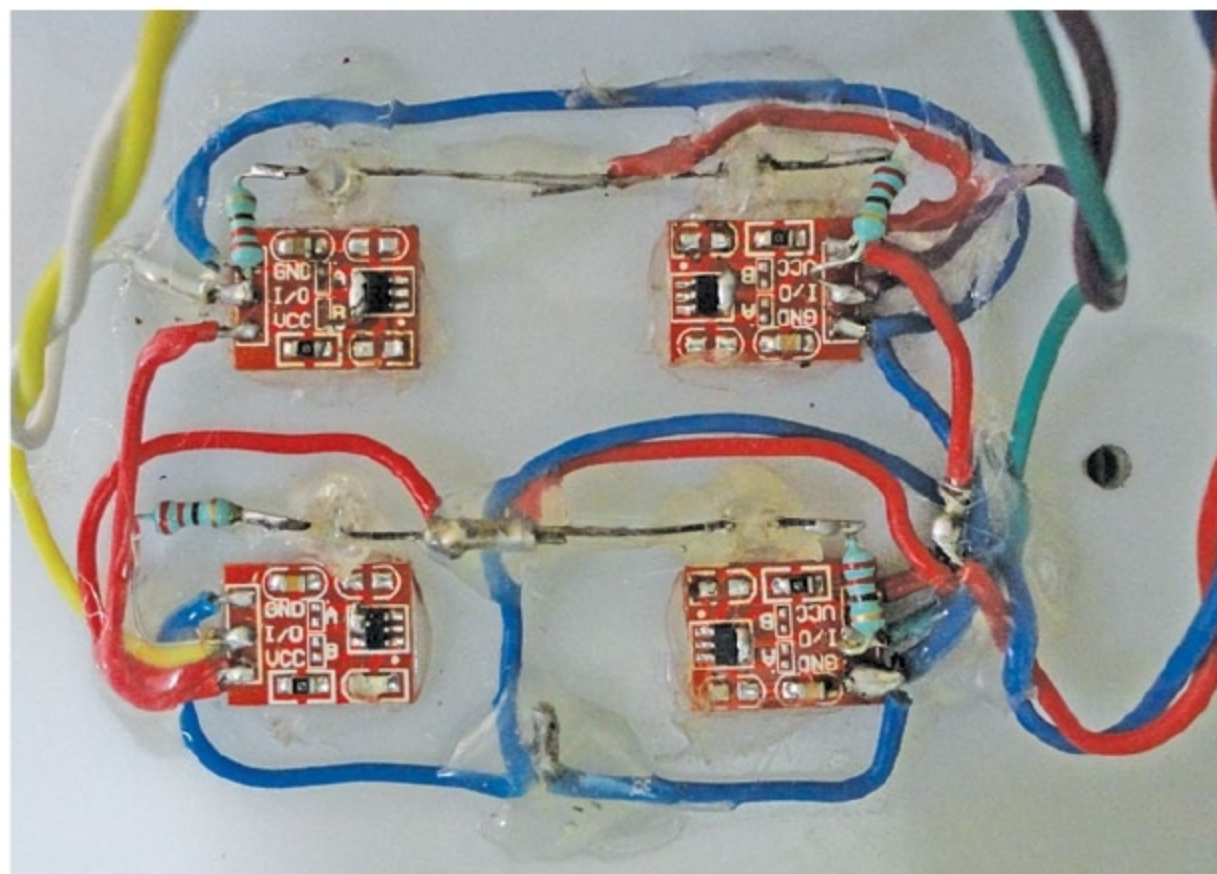


Fig. 7: Touchpad modules fixed on acrylic sheet

PARTS LIST

Semiconductors:

LED1-LED4 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R4 - 220-ohm

Miscellaneous:

TTP1-TTP4 - TTP223 module

CON1-CON4 - 3-pin connector

Module1 - 5V SMPS module

Module2 - 5V 4-channel SSR relay

B1-B3 - 230V AC, 60W bulb

BELL1 - 230V AC bell

- Box with acrylic sheet

(All these are available from kitsnspares.com)

supply is used here to ensure that all components are easily enclosed in the switch box. TTP223 IC can handle voltages between 2.0V to 5.5V, and power supply should not exceed 5.5V.

The circuit diagram of the four-channel touch switch panel is shown in Fig. 5. It is built around four inbuilt TTP223 modules (TTP1 through TTP4), 5V SMPS module (Module1), 5V four-channel SSR relay module (Module2), four LEDs (LED1 through LED4), three 230V AC, 60W bulbs (B1 through B3), 230V AC bell (BELL1) and a few other components.

Construction and testing

The relay board and 5V power supply module are suitably placed inside the switch box, as shown in Fig. 6. External lights and the bell are connected to output terminals of the relay board. Touchpads marked as I/O on the modules are connected to input pins

of relay modules.

The four modified touch modules along with four LEDs are placed on the acrylic sheet using hot glue, as shown on the right side of Fig. 6. Make four small holes for the LEDs (LED1 through LED4) on the front panel of acrylic sheet. LED1 through LED4 are fixed on the holes using a hot glue gun near the touch modules, such that when a touch module is touched, the corresponding LED glows (Fig. 7). LEDs should be properly placed in the holes so that these are clearly visible on the front panel.

After the connections are made, close the switch box and switch on the power supply. Label the three lights as 1, 2 and 3 on the front panel, as shown in the prototype (Fig. 4). Label BELL as doorbell for the fourth appliance. Now, touch any label, say, 1, on the front panel. LED1 will glow, and the corresponding relay will be energised, which will turn on the first light.

SSR relay module used in this project can handle maximum 2A current, and is suitable for room lights. For switching higher current appliances, replace SSR board with electro-mechanical relays that can handle currents up to 10A. **EFY**



Ashok Baijal is a regular contributor to EFY, currently retired and working on home automation projects

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